

6. Write a PHP program with a user interface for calculating age based on the user's birth date. Display the age in years, months, and days.**A6Age.php**

```
<!DOCTYPE html>

<html lang="en">

<head>

    <title>Age Calculator</title>

</head>

<body>

    <h2>Age Calculator</h2>

    <form method="post">

        <label for="birthdate">Enter your birth date:</label>

        <br>

        <input type="date" id="birthdate" name="birthdate">

        <br><br>

        <input type="submit" value="Calculate">

    </form>

    <?php

function calculateAge($birthdate){

    $today=new DateTime();

    $diff=$today->diff(new DateTime($birthdate));

    $years=$diff->y;

    $months=$diff->m;

    $days=$diff->d;

    return array($years,$months,$days);

}
```



```
if($_SERVER["REQUEST_METHOD"]=="POST"){

    $birthdate=$_POST["birthdate"];

    $todayDate=date("Y-m-d");

    if(!empty($birthdate)){

        if($birthdate>$todayDate){

            echo "<h3>Please enter a birth date that is not in the future.</h3>";

        }else{

            list($years,$months,$days)=calculateAge($birthdate);

            echo "<h3>Your age is: $years years, $months months,and $days days</h3>";

        }

    }else{

        echo "<h3>Please enter your birth date</h3>";

    } } ?>

</body>

</html>
```

OUTPUT:



Age Calculator

Enter your birth date:

dd - mm - yyyy 

Calculate

Please enter your birth date

Age Calculator

Enter your birth date:

02 - 10 - 1869 

Calculate

Your age is: 155 years, 4 months,and 26 days

Age Calculator

Enter your birth date:

07 - 07 - 2027 

Calculate

Please enter a birth date that is not in the future.

7. Write a PHP program that includes a user form with a text field and submit buttons for various string manipulations. It will display the result accordingly. (for replace, replace 'a' with 'x').

String Manipulation

Enter a string:

Shuffled string: niIda

```
<!DOCTYPE html>
```

```
<html> <head>
```

```
<title>String Manipulation</title>
```

```
</head> <body>
```

```
<h2>String Manipulation</h2>
```

```
<form method="post" action="<?php echo htmlspecialchars($_SERVER["PHP_SELF"]);?>">
```

```
    Enter a string: <input type="text" name="input_string" required>
```

```
    <br><br>
```

```
    <input type="submit" name="get_length" value="Get Length">
```

```
    <input type="submit" name="reverse" value="Reverse">
```

```
    <input type="submit" name="uppercase" value="Uppercase">
```

```
    <input type="submit" name="lowercase" value="Lowercase">
```

```
    <input type="submit" name="replace" value="Replace">
```

```
    <input type="submit" name="check_palindrome" value="Check Palindrome">
```

```
    <input type="submit" name="shuffle" value="Shuffle">
```

```
    <input type="submit" name="word_count" value="Word Count">
```

```
</form>
```

```
<?php
```

```
if($_SERVER["REQUEST_METHOD"]=="POST"){
```

```
    $input_string=$_POST["input_string"];
```

```
    if(isset($_POST["get_length"])){
```

```
        echo "Length of the string: ".strlen($input_string);
```

```
}

elseif(isset($_POST["reverse"]))

{

    echo "Reverse string: ".strrev($input_string);

}

elseif(isset($_POST["uppercase"]))

{

    echo "Uppercase string: ".strtoupper($input_string);

}

elseif(isset($_POST["lowercase"]))

{

    echo "Lowercase string: ".strtolower($input_string);

}

elseif(isset($_POST["replace"]))

{

    echo "Replaced string: ".str_replace('a','x',$input_string);

}

elseif (isset($_POST["check_palindrome"]))

{

    // Case-insensitive palindrome check

    $normalized_string = strtolower($input_string);

    $reversed_string = strrev($normalized_string);

    if ($normalized_string == $reversed_string)

    {

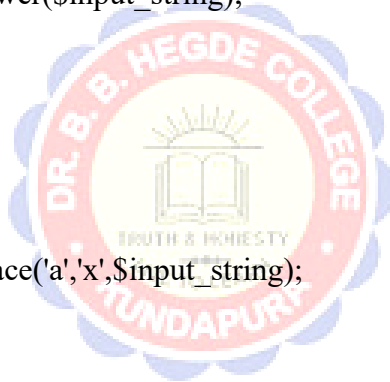
        echo "The string is a palindrome.";

    }

}

else {

    echo "The string is not a palindrome.";
```



```
}  
  
}  
  
elseif(isset($_POST["shuffle"])){  
    echo "Shuffled string: ".str_shuffle($input_string);  
}  
  
elseif(isset($_POST["word_count"])){  
    echo "Word count: ".str_word_count($input_string);  
}  
}  
?  
  
</body>  
</html>
```

OUTPUT:

String Length:

String Manipulation

Enter a string:

Length of the string: 14

8. Write a PHP user interface program with an HTML form to input a string. Upon submission, it will display the number of times each word occurs, ignoring the distinction between capital and lowercase letters. It should also print the most and least used words. Additionally, include buttons for sorting data in ascending order and descending order.

A8Sort.php

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
    <title>Word Frequency Analyzer</title>
```

```
</head>
```

```
<body>
```

```
<h2>Word Frequency Analyzer</h2>
```

```
<!-- Input Form -->
```

```
<form method="post">
```

```
    Enter a string:
```

```
    <input type="text" name="text" required
```

```
    value="<?= $_POST['text'] ?? " ?">">
```

```
    <button type="submit" name="analyze">Analyze</button>
```

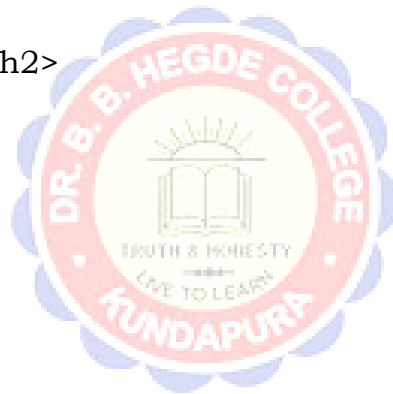
```
</form>
```

```
<?php
```

```
if ($_SERVER["REQUEST_METHOD"] == "POST") {
```

```
    // Step 1: Store original input
```

```
    $originalText = $_POST["text"];
```



```
// Step 2: Convert to lowercase ONLY for processing
```

```
$text = strtolower($originalText);
```

```
// Step 3: Split into words (handles multiple spaces)
```

```
$words = preg_split('/\s+/', trim($text));
```

```
// Step 4: Count frequency
```

```
$count = array_count_values($words);
```

```
// Step 5: Sorting
```

```
if (isset($_POST["asc"])) {
```

```
    asort($count);
```

```
} elseif (isset($_POST["desc"])) {
```

```
    arsort($count);
```

```
}
```

```
// Step 6: Display frequencies
```

```
echo "<h3>Word Frequencies:</h3>";
```

```
foreach ($count as $word => $c) {
```

```
    echo "$word: $c times<br>";
```

```
}
```

```
// Step 7: Find max & min
```

```
$max = max($count);
```

```
$min = min($count);
```

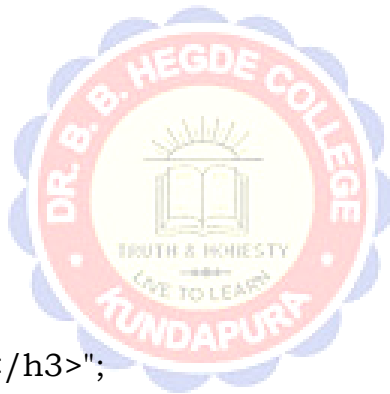
```
// Step 8: Get all most & least used words
```

```
$mostWords = array_keys($count, $max);
```

```
$leastWords = array_keys($count, $min);
```

```
// Convert to string
```

```
$most = implode(" ", $mostWords);
```



```

$least = implode(" ", $leastWords);

// Step 9: Display result

echo "<br><b>The most used word is: $most (used $max times)</b><br><br>";

echo "<b>The least used word is: $least (used $min times)</b>";

// Step 10: Sort buttons AFTER result (retain original input)

echo '

<form method="post">

    <input type="hidden" name="text" value="'. $originalText. '">

    <br><br>

    <button type="submit" name="asc">Sort Ascending</button><br><br>

    <button type="submit" name="desc">Sort Descending</button>

</form>    ' ; }

?>

</body>

</html>

```

OUTPUT

Word Frequency Analyzer

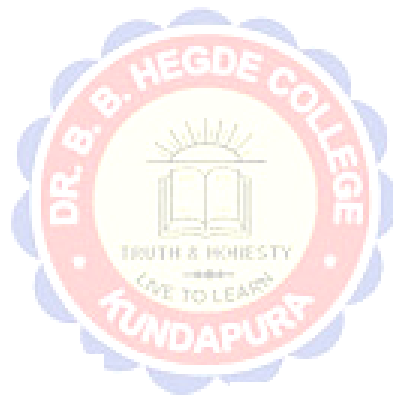
Enter a string:

Word Frequencies:

a: 1 times
b: 2 times
c: 1 times
d: 3 times
e: 4 times
f: 5 times

The most used word is: f (used 5 times)

The least used word is: a, c (used 1 times)



Word Frequency Analyzer

Enter a string:

Word Frequencies:

f: 5 times
e: 4 times
d: 3 times
b: 2 times
a: 1 times
c: 1 times

The most used word is: f (used 5 times)

The least used word is: a, c (used 1 times)

1.PHP program to implement student registration form using Labels, Text Boxes, Text Area, Checkbox, Radio Buttons, Select and Submit button. (First Name, Last Name, Address, E-Mail, Mobile, City, State, Gender, Hobbies, Blood Group). Display user inserted value in a new PHP page in a neat format.

B1.php

```
<!DOCTYPE html>

<html lang="en">

<head>  <meta charset="UTF-8">

        <meta name="viewport" content="width=device-width, initial-scale=1.0">

        <title>Student Registration Form</title> </head>

<body>  <h2>Student Registration Form</h2>

        <form action="b1Process.php" method="post">

            First Name: <input type="text" name="firstname" id="firstname" required><br><br>

            Last Name: <input type="text" name="lastname" id="lastname" required><br><br>

            <div style="display:flex; align-items:center;">

                Address:

                <textarea name="address" id="address" rows="4" cols="30" required></textarea>

            </div><br><br>

            Email: <input type="email" name="email" id="email"><br><br>

            Mobile:

            <input type="tel" name="mobile" id="mobile" pattern="[0-9]{10}" title="Please enter a 10
            digit mobile number" required><br><br>

            City:  <input type="text" name="city" id="city" required><br><br>

            State: <input type="text" name="state" id="state" required><br><br>

            Gender:<br>

            <input type="radio" name="gender" value="Male" id="male" required> Male<br>

            <input type="radio" name="gender" value="Female" id="female" required> Female<br><br>
```

Hobbies:

<input type="checkbox" name="hobbies[]" id="reading" value="Reading">Reading:

<input type="checkbox" name="hobbies[]" id="sports" value="Sports"> Sports:

<input type="checkbox" name="hobbies[]" id="music" value="Music"> Music:

<input type="checkbox" name="hobbies[]" id="other" value="other">

Other:

Blood group: <select name="bloodgroup" id="bloodgroup" required>

<option value="">Select</option>

<option value="A+">A+</option>

<option value="A-">A-</option>

<option value="B+">B+</option>

<option value="B-">B-</option>

<option value="AB+">AB+</option>

<option value="AB-">AB-</option>

<option value="O+">O+</option>

<option value="O-">O-</option>

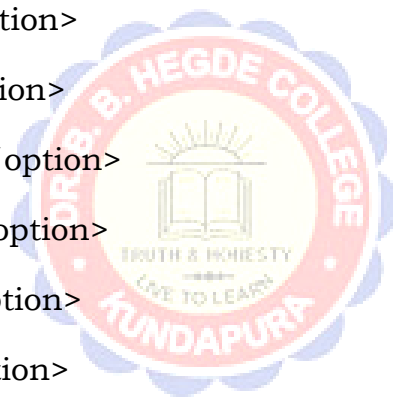
</select>

<input type="submit" Value="Submit">

</form>

</body>

</html>



```
<!DOCTYPE html>

<html lang="en">

<head>  <meta charset="UTF-8">

        <meta name="viewport" content="width=device-width, initial-scale=1.0">

        <title>Document</title>  </head>

<body>

<?php

if($_SERVER["REQUEST_METHOD"]=="POST")

{

    echo "<p><strong>First Name: </strong>" . $_POST['firstname'] . "</p>";

    echo "<p><strong>Last Name: </strong>" . $_POST['lastname'] . "</p>";

    echo "<p><strong>Address: </strong>" . $_POST['address'] . "</p>";

    echo "<p><strong>Email: </strong>" . $_POST['email'] . "</p>";

    echo "<p><strong>Mobile: </strong>" . $_POST['mobile'] . "</p>";

    echo "<p><strong>City: </strong>" . $_POST['city'] . "</p>";

    echo "<p><strong>State: </strong>" . $_POST['state'] . "</p>";

    echo "<p><strong>Gender: </strong>" . $_POST['gender'] . "</p>";

    echo "<p><strong>Hobbies: </strong>".implode(", ", $_POST['hobbies']) . "</p>";

    echo "<p><strong>Blood group: </strong>" . $_POST['bloodgroup'] . "</p>";

}

?>

</body>

</html>
```

OUTPUT

Student F x (167) Wh x +

localhost/b1.php

Student Registration Form

First Name:

Last Name:

Address:

Email:

Mobile:

City:

State:

Gender:
☐ Male
☐ Female

Hobbies:
☐ Reading:
☐ Sports:
☐ Music:
☐ Other:

Blood group:

Student F x (167) Wh x +

localhost/b1.php

Student Registration Form

First Name:

Last Name:

Address:

Email:

Mobile:

City:

State:

Gender:
☒ Male
☐ Female

Hobbies:
☒ Reading:
☐ Sports:
☒ Music:
☐ Other:

Blood group:



The image shows a web browser window with two tabs. The active tab is titled 'Document' and the address bar shows 'localhost/b1Proces...'. The page content displays a form with the following fields:

First Name: Jyothiradithya

Last Name: Bhat

Address: Near Sharon Hotel Kundapura

Email: bhatjyothi@gmail.com

Mobile: 9844611765

City: Udupi

State: Karnataka

Gender: Male

Hobbies: Reading, Music

Blood group: B+

2. Develop a PHP program that facilitates the addition, multiplication of two matrices. Utilize HTML for the user interface and PHP for the backend logic. Dynamically generate the required number of textboxes based on the specified number of rows and columns. Implement three distinct buttons to perform each matrix operation For instance, consider the addition of two matrices as an example.

Code:

PB2Matrix.php

```
<!DOCTYPE html><html><head><title>Matrix Ops</title></head><body>

<h2>Matrix Operations</h2>

<form method="post">

    M1: Rows <input type="number" name="r1" min="1" required value="<?=$_POST['r1'] ?? " ?>">

    Cols <input type="number" name="c1" min="1" required value="<?=$_POST['c1'] ?? " ?>"><br><br>

    M2: Rows <input type="number" name="r2" min="1" required value="<?=$_POST['r2'] ?? " ?>">

    Cols <input type="number" name="c2" min="1" required value="<?=$_POST['c2'] ?? " ?>"><br><br>

    <button name="op" value="add">Add</button>

    <button name="op" value="mul">Multiply</button>

</form>
```

```
<?php

function matrixInput($r, $c, $name, $data = []) {

    echo "<h4>$name</h4><table>";

    for ($i = 0; $i < $r; $i++) {

        echo "<tr>";

        for ($j = 0; $j < $c; $j++) {

            $v = $data[$i][$j] ?? "";

            echo "<td><input type='number' name='{ $name }[$i][$j]' value='$v' required></td>";

        }

        echo "</tr>";

    }

}
```

```
}

echo "</table><br>";

}

function matrixOutput($m, $title) {

    echo "<h3>$title</h3><table border=1>";

    foreach ($m as $r) {

        echo "<tr>";

        foreach ($r as $v) echo "<td>$v</td>";

        echo "</tr>";

    }

    echo "</table><br>";

}

if (isset($_POST['op'])) {

    [$r1,$c1,$r2,$c2,$op] = [$_POST['r1'],$_POST['c1'],$_POST['r2'],$_POST['c2'],$_POST['op']];

    $valid = ($op=="add" && $r1==$r2 && $c1==$c2) || ($op=="mul" && $c1==$r2);

    if ($valid) {

        echo "<form method='post'>";

        foreach (['r1','c1','r2','c2','op'] as $f) echo "<input type='hidden' name='$f' value='{$_POST[$f]}'>";

        matrixInput($r1, $c1, "m1", $_POST['m1'] ?? []);

        matrixInput($r2, $c2, "m2", $_POST['m2'] ?? []);

        echo "<button name='calc'>Calculate</button></form>";

    } else echo "<h3> Invalid matrix dimensions</h3>";

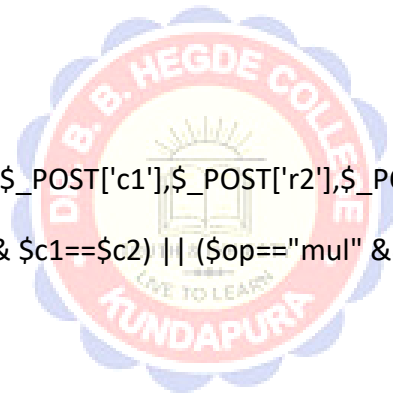
}

if (isset($_POST['calc'])) {

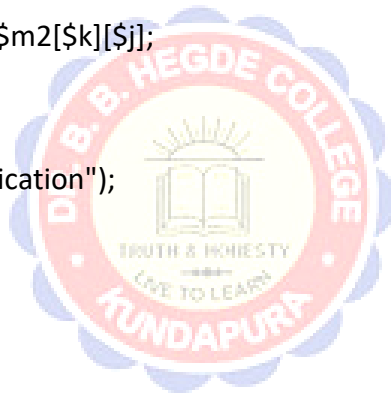
    [$m1, $m2, $op] = [$_POST['m1'], $_POST['m2'], $_POST['op']];

    $r1 = count($m1); $c1 = count($m1[0]); $c2 = count($m2[0]);

    $res = [];
```



```
if ($op == "add") {  
    for ($i=0; $i<$r1; $i++)  
        for ($j=0; $j<$c1; $j++)  
            $res[$i][$j] = $m1[$i][$j] + $m2[$i][$j];  
    matrixOutput($res, "Result: Addition");  
}  
  
if ($op == "mul") {  
    for ($i=0; $i<$r1; $i++)  
        for ($j=0; $j<$c2; $j++) {  
            $res[$i][$j] = 0;  
            for ($k=0; $k<$c1; $k++)  
                $res[$i][$j] += $m1[$i][$k] * $m2[$k][$j];  
        }  
    matrixOutput($res, "Result: Multiplication");  
}  
}  
?>  
  
</body>  
  
</html>
```



Output:

Matrix Operations

M1: Rows Cols M2: Rows Cols

Matrix addition

Matrix Operations

M1: Rows Cols M2: Rows Cols **m1**

1	2
3	4

m2

4	3
2	1

Result: Addition

5	5
5	5

Matrix Multiplication

Matrix Operations

M1: Rows Cols M2: Rows Cols **m1**

1	2
3	4

m2

4	3
2	1

Result: Multiplication

8	5
20	13

3. Write a PHP program that implements a class to add and find the difference of two distance values given in feet and inches. The user inputs are collected through an HTML form.

```
<!DOCTYPE html>

<html>

<head>  <title>Distance Calculator</title></head>

<body> <h2>Distance Calculator</h2>

<form method="post">

  <p>Distance 1:</p>

  Feet: <input type="number" name="feet1" min="0" required><br><br>

  Inches: <input type="number" name="inches1" min="0" max="11" required><br><br>

  <p>Distance 2:</p>

  Feet: <input type="number" name="feet2" min="0" required><br><br>

  Inches: <input type="number" name="inches2" min="0" max="11" required><br><br>

  <input type="submit" value="Calculate">

</form>

<?php

class DistanceCalculator {

  function add($f1, $i1, $f2, $i2) {

    $total = ($f1*12 + $i1) + ($f2*12 + $i2);

    return [floor($total/12), $total%12];

  }

  function diff($f1, $i1, $f2, $i2) {

    $total = abs(($f1*12 + $i1) - ($f2*12 + $i2));

    return [floor($total/12), $total%12];

  }

}
```

```
if ($_SERVER["REQUEST_METHOD"] == "POST") {

    $f1 = $_POST['feet1'];
    $i1 = $_POST['inches1'];
    $f2 = $_POST['feet2'];
    $i2 = $_POST['inches2'];

    // Validation
    if ($f1 < 0 || $f2 < 0 || $i1 < 0 || $i2 < 0 || $i1 > 11 || $i2 > 11) {
        echo "<p style='color:red;'>Invalid input! Inches must be 0–11 and values non-
negative.</p>";
    }
else
{
    $obj = new DistanceCalculator();
    list($sf, $si) = $obj->add($f1,$i1,$f2,$i2);
    list($df, $di) = $obj->diff($f1,$i1,$f2,$i2);
    echo "<h3>Result:</h3>";
    echo "<b>Distance 1:</b> $f1' $i1" <br>";
    echo "<b>Distance 2:</b> $f2' $i2" <br><br>";
    echo "<b>Sum:</b> $sf $si" <br>";
    echo "<b>Difference:</b> $df $di"";
}
}
?>

</body> </html>
```

OUTPUT

Distance Calculator

Distance 1:

Feet:

Inches:

Distance 2:

Feet:

Inches:

Distance Calculator

Distance 1:

Feet:

Inches:

Distance 2:

Feet:

Inches:

Distance Calculator

Distance 1:

Feet:

Inches:

Distance 2:

Feet:

Inches:

Result:

Distance 1: 10' 10"

Distance 2: 5' 2"

Sum: 16' 0"

Difference: 5' 8"

4. Program to implement a login form where users enter their username and password. Validate the credentials against data stored in a MySQL database and grant access if they are correct.

Database name: BBHC

Database structure:

Table name: users

#	Name	Type
☐ 1	UserName 	varchar(20)
☐ 2	Password	varchar(20)

B4Login.php Code:

```
<?php
session_start();

// Database connection
$servername="localhost";
$username="root";
$password="";
$dbase="bbhc";
$port=3306;

$conn=mysqli_connect($servername,$username,$password,$dbase,$port);

if(!$conn)
{
die("Connection failed: ".mysqli_connect_error());
}

// If already logged in
if(isset($_SESSION['username']))
{
header("Location:B4Welcome.php");
exit();
}
```

// Check form submission

```
if(isset($_POST['login']))  
{  
    $username=$_POST['username'];  
    $password=$_POST['password'];  
    $sql="SELECT * FROM users WHERE username='$username' AND password='$password'  
";  
    $result=mysqli_query($conn,$sql);  
  
    if(mysqli_num_rows($result)>0)  
    {  
        $_SESSION['username']=$username;  
        header("Location:B4Welcome.php");  
        exit();  
    }  
    else  
    {  
        echo "<p style='color:red'>Invalid Username or Password</p>";  
    }  
}
```

<html>

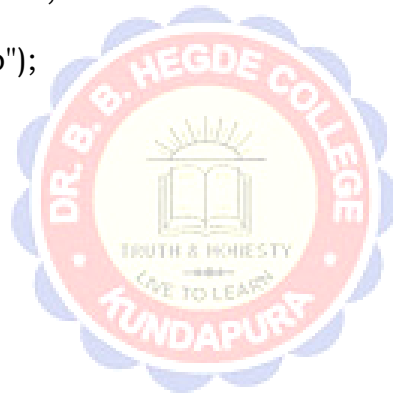
<head>

<title>Login</title>

</head>

<body>

<h2>Login Form</h2>



```
<form method="post">
```

```
Username:<br>
```

```
<input type="text" name="username" required><br><br>
```

```
Password:<br>
```

```
<input type="password" name="password" required><br><br>
```

```
<input type="submit" name="login" value="Login">
```

```
</form>
```

```
</body>
```

```
</html>
```

B4Welcome.php

```
<?php
```

```
session_start();
```

```
// Check whether user is logged in
```

```
if(!isset($_SESSION['username']))
```

```
{
```

```
    header("Location:B4Login.php");
```

```
    exit();
```

```
}
```

```
// Logout
```

```
if(isset($_GET['logout']) && $_GET['logout']==1)
```

```
{
```

```
    session_destroy();
```

```
    header("Location:B4Login.php");
```

```
    exit();
```

```
}
```

```
?>
```



```
<html>
<head>
<title>Welcome</title>
</head>
<body>
<h2>Successfully Logged In</h2>
Welcome <?php echo $_SESSION['username']; ?>
<br><br>
<a href="B4welcome.php?logout=1">Logout</a>
</body>
</html>
```

OUTPUT

Login Form

Username:

Password:

Login



Login Form

Username:

Password:

Login

Successfully Logged In

Welcome Anvith

[Logout](#)

5. Program to create a feedback form with fields for name, email, subject, and message. Store the submitted feedback data in a MySQL database for later review.

Database : BBHC

Table : Feedback

#	Name	Type	Collation	Attributes	Null	Default
1	Name	varchar(22)	utf8mb4_general_ci		No	None
2	Email	varchar(22)	utf8mb4_general_ci		No	None
3	Subject	varchar(22)	utf8mb4_general_ci		No	None
4	Msg	varchar(22)	utf8mb4_general_ci		No	None
5	Time	timestamp			No	current_timestamp()

B5Feedback.php

```
<?php
```

```
$servername = "localhost";
```

```
$UserName = "root";
```

```
$Password = "";
```

```
$dbname = "BBHC";
```

```
$port = 3306;
```

```
$conn = new mysqli($servername, $UserName, $Password, $dbname, $port);
```

```
if ($conn->connect_error)
```

```
{
```

```
    die("Connection failed: " . $conn->connect_error);
```

```
}
```

```
if ($_SERVER["REQUEST_METHOD"] == "POST")
```

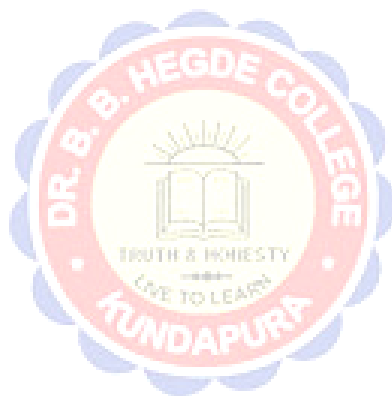
```
{
```

```
    $name = $_POST['name'];
```

```
    $email = $_POST['email'];
```

```
    $subject = $_POST['subject'];
```

```
    $msg = $_POST['msg'];
```



```
$sql = "INSERT INTO feedback (Name, Email, Subject, Msg)
```

```
VALUES ('$name','$email','$subject','$msg')";
```

```
if ($conn->query($sql) === TRUE)
```

```
{
```

```
echo "Feedback Submitted Successfully<br><br>";
```

```
echo "<a href='?view=feedback'>View Latest Feedback</a><br><br>";
```

```
echo "<a href=''>Back to Feedback Page</a>";
```

```
}
```

```
else
```

```
{
```

```
echo "Error: " . $conn->error;
```

```
}
```

```
}
```

```
else
```

```
{
```

```
?>
```

```
<!DOCTYPE html>
```

```
<html> <head> <title>Feedback Form</title> </head>
```

```
<body> <h2>Feedback Form</h2>
```

```
<form method="post">
```

```
Name: <input type="text" name="name" required><br><br>
```

```
Email: <input type="email" name="email" required><br><br>
```

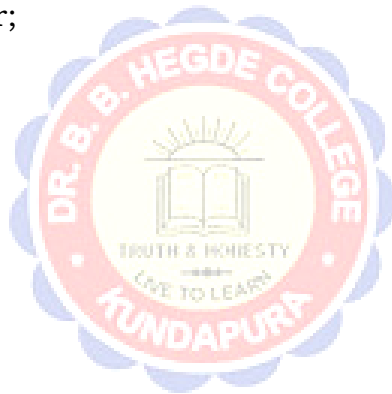
```
Subject: <input type="text" name="subject" required> <br><br>
```

```
Message: <textarea name="msg" rows="4" required></textarea><br><br>
```

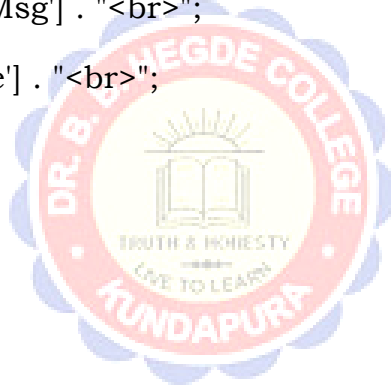
```
<input type="submit" value="Submit Feedback"> </form>
```

```
<?php
```

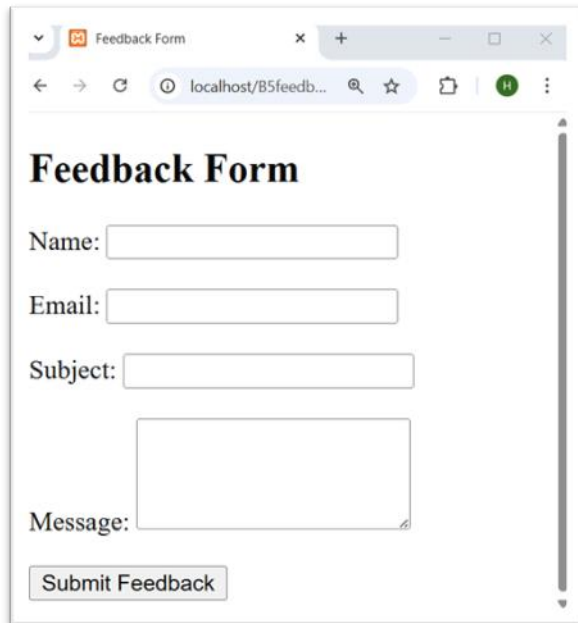
```
if (isset($_GET['view']) && $_GET['view'] == 'feedback') {
```



```
$sql = "SELECT * FROM feedback ORDER BY Time DESC LIMIT 1";  
$result = $conn->query($sql);  
if ($result->num_rows > 0)  
{  
    echo "<h3>Latest Feedback Submitted</h3>";  
    while ($row = $result->fetch_assoc())  
    {  
        echo "Name: " . $row['Name'] . "<br>";  
        echo "Email: " . $row['Email'] . "<br>";  
        echo "Subject: " . $row['Subject'] . "<br>";  
        echo "Message: " . $row['Msg'] . "<br>";  
        echo "Time: " . $row['Time'] . "<br>";  
    }  
}  
else  
{  
    echo "No feedback found.";  
}  
}  
?>  
</body> </html>  
<?php  
}  
$conn->close(); ?>
```



OUTPUT



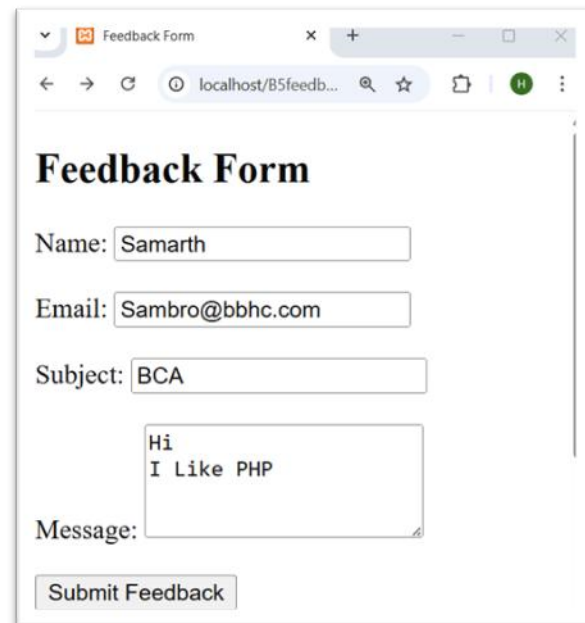
Feedback Form

Name:

Email:

Subject:

Message:



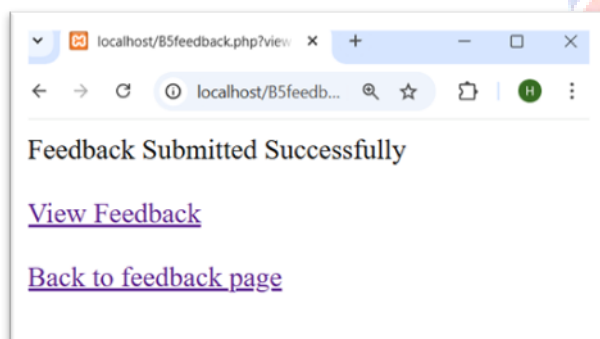
Feedback Form

Name:

Email:

Subject:

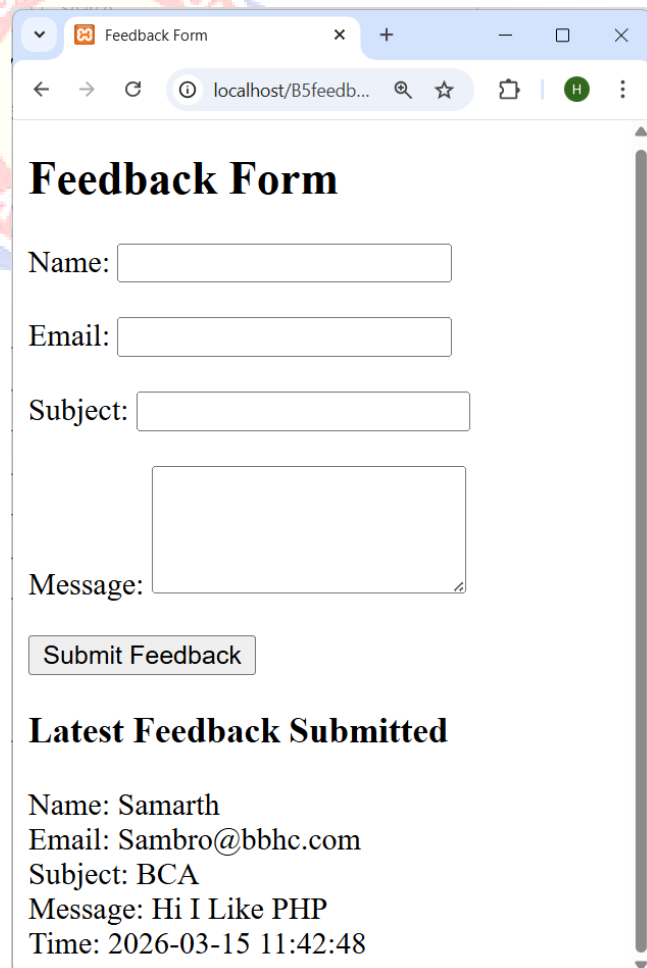
Message:

[View Feedback](#)

Feedback Submitted Successfully

[View Feedback](#)

[Back to feedback page](#)



Feedback Form

Name:

Email:

Subject:

Message:

Latest Feedback Submitted

Name: Samarth
Email: Sambro@bbhc.com
Subject: BCA
Message: Hi I Like PHP
Time: 2026-03-15 11:42:48

6. Develop a dynamic PHP application to efficiently manage and store customer information, encompassing key fields like Customer Number, Customer Name, Item Purchased, and Mobile Number in Database. This application should provide a user friendly interface with strategically placed buttons to trigger specific functionalities. These functionalities include:

1. Add Customer Information: Clicking this button should dynamically reveal a form for entering new customer details. Include proper validation checks for mobile numbers (10 digits), and also for Customer id ensuring accuracy in data input.
2. Delete Customer Records: Triggering this button should prompt the appearance of a form, specifically requesting the Customer ID to identify and delete the corresponding customer record. And provide appropriate messages for incorrect inputs.
3. Search for Particular Entries: This function should unveil a search form when activated, allowing users to input Customer id to find specific customer records.
4. Sort Database Based on Customer Id: Clicking this button should facilitate the sorting of the entire database based on customer id.
5. Display Complete Set of Records: Activating this function should present a comprehensive display of all customer records.
6. Interface Design: Initially, the interface should only showcase functional buttons. Upon clicking a button, the respective form should dynamically appear, offering a tailored and focused user experience.
7. Give proper messages after every transaction.

Table Name: Customers

CREATE TABLE customers (

id INT(11) NOT NULL AUTO_INCREMENT,
customer_id INT(11) NOT NULL,
customer_name VARCHAR(50) NOT NULL,
item_purchased VARCHAR(50) NOT NULL,
mobile_number VARCHAR(15) NOT NULL,
PRIMARY KEY (id)

);

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
☐ 1	id 	int(11)			No	None		AUTO_INCREMENT
☐ 2	customer_id 	int(11)			No	None		
☐ 3	customer_name	varchar(50)	utf8mb4_general_ci		No	None		
☐ 4	item_purchased	varchar(50)	utf8mb4_general_ci		No	None		
☐ 5	mobile_number	varchar(15)	utf8mb4_general_ci		No	None		

```
<?php
```

```
$conn = mysqli_connect("localhost","root","","bbhc", "3306") or die("Connection failed");
```

```
$msg = "";
```

```
$res = null;
```

```
// --- ADD ---
```

```
if(isset($_POST['add'])){\
```

```
    $id = $_POST['customer_id'];
```

```
    $chk = mysqli_query($conn,"SELECT * FROM customers WHERE customer_id=$id");
```

```
    if(mysqli_num_rows($chk) > 0){
```

```
        $msg = "Customer ID already exists";
```

```
    } else {
```

```
        mysqli_query($conn,"INSERT INTO customers(customer_id, customer_name,  
item_purchased, mobile_number)
```

```
            VALUES($id,$_POST[name'],$_POST[item'],$_POST[mob])");
```

```
        $msg = "Customer Added";
```

```
    } }
```

```
// --- DELETE ---
```

```
if(isset($_POST['delete'])){\
```

```
    mysqli_query($conn,"DELETE FROM customers WHERE  
customer_id=$_POST[customer_id]");
```

```
    $msg = mysqli_affected_rows($conn) > 0 ? "Customer Deleted" : "No such ID";
```

```
}
```

```
// --- SEARCH ---
```

```
if(isset($_POST['search'])){\
```

```
    $res = mysqli_query($conn,"SELECT * FROM customers WHERE  
customer_id=$_POST[customer_id]");
```

```
    if(mysqli_num_rows($res) == 0){
```

```
        $msg = "No Record Found";
```

```
$res = null;
```

```
} }
```

```
// --- DISPLAY (in insertion order) ---
```

```
if(isset($_POST['display'])){\
```

```
    $res = mysqli_query($conn,"SELECT * FROM customers ORDER BY id"); // insertion  
    order
```

```
    if(mysqli_num_rows($res) == 0){
```

```
        $msg = "No Records Found";
```

```
        $res = null;
```

```
    } }
```

```
// --- SORT (by customer_id) ---
```

```
if(isset($_POST['sort'])){\
```

```
    $res = mysqli_query($conn,"SELECT * FROM customers ORDER BY customer_id");
```

```
    if(mysqli_num_rows($res) == 0){
```

```
        $msg = "No Records Found";
```

```
        $res = null;
```

```
    } } ?>
```

```
<html>
```

```
<head>
```

```
<title>Customer Management System</title>
```

```
<script>
```

```
function show(id){
```

```
    ['addDiv','delDiv','searchDiv'].forEach(x =>
```

```
    document.getElementById(x).style.display='none');
```

```
    document.getElementById(id).style.display='block';
```

```
}
```

```
</script>
```

```
</head>
```



```
<body>
```

```
<h3>Customer Management System</h3>
```

```
<p style="color:green;"><b><?= $msg ?></b></p>
```

```
<!-- Buttons -->
```

```
<button onclick="show('addDiv')">Add</button>
```

```
<button onclick="show('delDiv')">Delete</button>
```

```
<button onclick="show('searchDiv')">Search</button>
```

```
<form method="post" style="display:inline;">
```

```
    <button name="sort">Sort</button>
```

```
    <button name="display">Display</button>
```

```
</form> <br><br>
```

```
<!-- Table Output -->
```

```
<?php if($res){ ?>
```

```
<table border="1" cellpadding="8">
```

```
<tr><th>Customer ID</th><th>Name</th><th>Item</th><th>Mobile</th></tr>
```

```
<?php while($r = mysqli_fetch_assoc($res)){ ?>
```

```
<tr>
```

```
<td><?= $r['customer_id'] ?></td>
```

```
<td><?= $r['customer_name'] ?></td>
```

```
<td><?= $r['item_purchased'] ?></td>
```

```
<td><?= $r['mobile_number'] ?></td>
```

```
</tr>
```

```
<?php } ?>
```

```
</table>
```

```
<?php } ?>
```

```
<br>
```



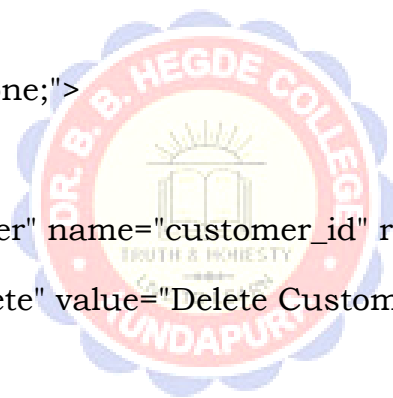

```
<div id="addDiv" style="display:none;">
<form method="post">
Customer ID:<input type="number" name="customer_id" required><br>
Name:<input type="text" name="name" required><br>
Item:<input type="text" name="item" required><br>
Mobile:<input type="text" name="mob" pattern="[0-9]{10}" required><br>
<input type="submit" name="add" value="Add Customer">
</form>
</div>
```

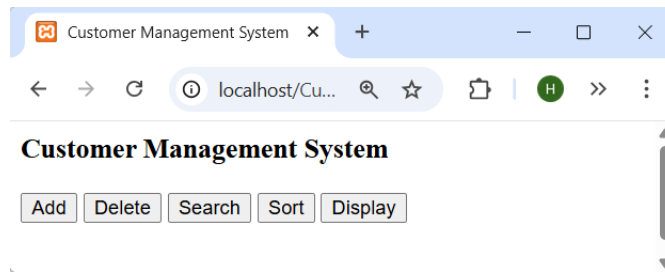
<!-- DELETE Form -->

```
<div id="delDiv" style="display:none;">
<form method="post">
Customer ID:<input type="number" name="customer_id" required><br>
<input type="submit" name="delete" value="Delete Customer">
</form>
</div>
```

<!-- SEARCH Form -->

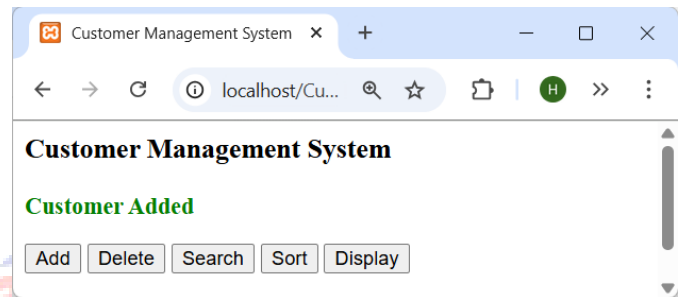
```
<div id="searchDiv" style="display:none;">
<form method="post">
Customer ID:<input type="number" name="customer_id" required><br>
<input type="submit" name="search" value="Search Customer">
</form> </div>
</body> </html>
```





Add Record

The screenshot shows the 'Add Record' form in the 'Customer Management System'. It includes input fields for 'Customer ID' (with value 80), 'Name' (with value Vignesh), 'Item' (with value Pen), and 'Mobile' (with value 7411312759). There is an 'Add Customer' button at the bottom.



Display Record

The screenshot shows the 'Display Record' table in the 'Customer Management System'. The table has four columns: 'Customer ID', 'Name', 'Item', and 'Mobile'. It contains eight rows of data.

Customer ID	Name	Item	Mobile
111	Samarth	Book	9844611765
99	Nishmitha	Pen	9844611765
80	Vignesh	Pen	7411312759
105	Aditya	Laptop	9353839699
5	Sameeksha	Book	7894561230
786	Abkari Sayeem	Scanner	9632154870
530	Thrisha C	Notes	4569871237

Sort Record

Customer Management System

Add Delete Search Sort Display

Customer ID	Name	Item	Mobile
5	Sameeksha	Book	7894561230
80	Vignesh	Pen	7411312759
99	Nishmitha	Pen	9844611765
105	Aditya	Laptop	9353839699
111	Samarth	Book	9844611765
530	Thrisha C	Notes	4569871237
786	Abkari Sayeem	Scanner	9632154870

Search Record

Customer Management System

Add Delete Search Sort Display

Customer ID	Name	Item	Mobile
111	Samarth	Book	9844611765

Customer ID:

Customer Management System

No Record Found

Add Delete Search Sort Display

Customer ID:

Delete Record

Customer ID	Name	Item	Mobile
111	Samarth	Book	9844611765
99	Nishmitha	Pen	9844611765
80	Vignesh	Pen	7411312759
105	Aditya	Laptop	9353839699
5	Sameeksha	Book	7894561230
786	Abkari Sayeem	Scanner	9632154870
530	Thrisha C	Notes	4569871237

Customer Management System

Customer Deleted

Customer ID:

Customer Management System

Customer ID	Name	Item	Mobile
111	Samarth	Book	9844611765
80	Vignesh	Pen	7411312759
105	Aditya	Laptop	9353839699
5	Sameeksha	Book	7894561230
786	Abkari Sayeem	Scanner	9632154870
530	Thrisha C	Notes	4569871237

Customer Management System

No such ID

Customer ID:

7. A PHP and MySQL programme that features a book shopping form that takes in the book number, book title, price, quantity, and a option to choose the book code. The bill with the discounted amount and net bill amount is then displayed. Additionally, bill data are stored in the table.

Code Discount rate

101 15%

102 20%

103 25%

Any other 5%

Find the discount amount and Net bill amount. Display the bill.

B7bookshop.php

```
<?php
```

```
$host = "localhost";
```

```
$username = "root";
```

```
$password = "";
```

```
$database = "bbhc";
```

```
$port=3307;
```

```
Mysqli_report(MYSQLI_REPORT_ERROR| MYSQLI_REPORT_STRICT);
```

```
$conn = mysqli_connect($host, $username, $password, $database,$port);
```

```
if (!$conn) { die("Connection failed: " . mysqli_connect_error()); }
```

```
// Initialize variables
```

```
$bookno = "";
```

```
$title = "";
```

```
$price = "";
```

```
$qty = "";
```

```
$code = 0;
```

```
$discount = 0;
```

```
$net = 0;
```

Table name : bookshop

#	Name	Type	Collation	Attributes	Null	Default
☐ 1	bookno 	int(11)			No	None
☐ 2	title	varchar(100)	utf8mb4_general_ci		Yes	NULL
☐ 3	price	double			Yes	NULL
☐ 4	qty	int(11)			Yes	NULL
☐ 5	code	int(11)			Yes	NULL
☐ 6	discount	double			Yes	NULL
☐ 7	net	double			Yes	NULL

```
$total = 0;
```

```
$message = "";
```

```
$showBill = false; // Flag to control whether to display bill
```

```
if (isset($_POST['submit'])) {
```

```
    // Get and convert input values
```

```
    $bookno = (int) $_POST['bookno'];
```

```
    $title = $_POST['title'];
```

```
    $price = (float) $_POST['price'];
```

```
    $qty = (int) $_POST['qty'];
```

```
    $code = (int) $_POST['code'];
```

```
    // Calculate total and discount
```

```
    $total = $price * $qty;
```

```
    if ($code == 101) {
```

```
        $discount = $total * 0.15;
```

```
    } elseif ($code == 102) {
```

```
        $discount = $total * 0.20;
```

```
    } elseif ($code == 103) {
```

```
        $discount = $total * 0.25;
```

```
    } else { $discount = $total * 0.05; }
```

```
    $net = $total - $discount;
```

```
    // Try to insert record
```

```
    try {
```

```
        $stmt = $conn->prepare("INSERT INTO bookshop (bookno, title, price, qty, code, discount, net) VALUES (?, ?, ?, ?, ?, ?, ?)");
```

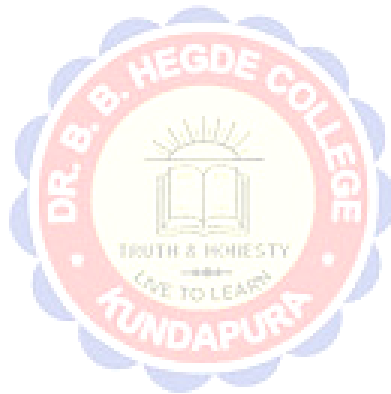
```
        $stmt->bind_param("isdiidd", $bookno, $title, $price, $qty, $code, $discount, $net);
```

```
        $stmt->execute();
```

```
        $message = "<span style='color:green;'>Record inserted successfully!</span>";
```

```
        $showBill = true; // Only show bill if insertion succeeded
```

```
    } catch (mysqli_sql_exception $e) {
```



```
if ($e->getCode() == 1062) {

    $message = "<span style='color:red;'>Error: Book Number already exists. Please use a different Book
Number.</span>";

    $showBill = false; // Don't show calculation

} else {

    $message = "<span style='color:red;'>Error: " . $e->getMessage() . "</span>";

    $showBill = false;

}

}

?>

<html>

<body>

<h2>Book Shopping Form</h2>

<form method="post">

Book Number :<input type="text" name="bookno" value="<?php echo $bookno; ?>" required><br><br>

Book Title :<input type="text" name="title" value="<?php echo $title; ?>" required><br><br>

Price : <input type="number" name="price" min="1" value="<?php echo $price; ?>" required><br><br>

Quantity :<input type="number" name="qty" min="1" value="<?php echo $qty; ?>" required><br><br>

Book Code : <select name="code">

<option value="101" <?php if($code==101) echo "selected"; ?>>101</option>

<option value="102" <?php if($code==102) echo "selected"; ?>>102</option>

<option value="103" <?php if($code==103) echo "selected"; ?>>103</option>

<option value="0" <?php if($code==0) echo "selected"; ?>>Other</option> </select> <br><br>

<input type="submit" name="submit" value="Calculate Bill">

</form> <br><br>

<?php

if ($showBill)

{
```

```
echo "<h3>Bill Details</h3>";

echo "-----<br>";

echo "Book Number : " . $bookno . "<br>";

echo "Book Title : " . $title . "<br>";

echo "Price : " . $price . "<br>";

echo "Quantity : " . $qty . "<br>";

echo "Total Amount : " . $total . "<br>";

echo "Code : " . $code . "<br>";

echo "Discount Amount : " . $discount . "<br>";

echo "Net Bill Amount : " . $net . "<br>";

}

if (!empty($message))
{
    echo "<br><b>$message</b>";
}

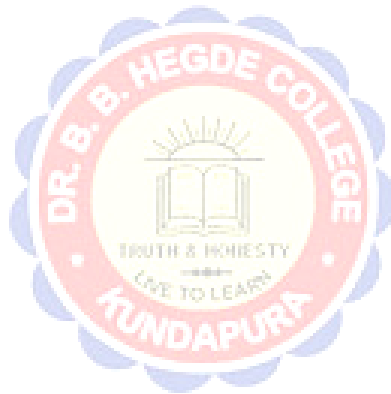
?>

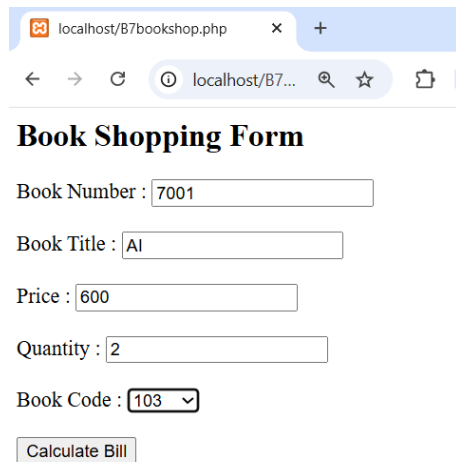
<br><br>

<a href="B7Bookshop.php">Go Back</a>

</body>

</html>
```

OUTPUT



localhost/B7bookshop.php

Book Shopping Form

Book Number : 7001

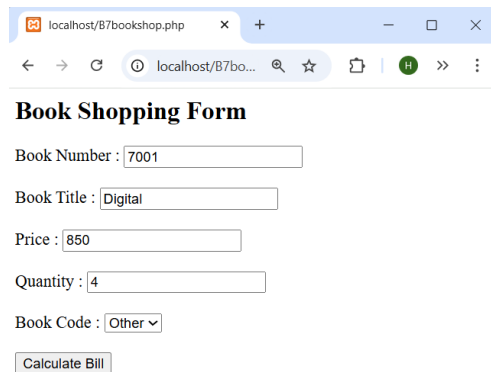
Book Title : AI

Price : 600

Quantity : 2

Book Code : 103

Calculate Bill



localhost/B7bookshop.php

Book Shopping Form

Book Number : 7001

Book Title : Digital

Price : 850

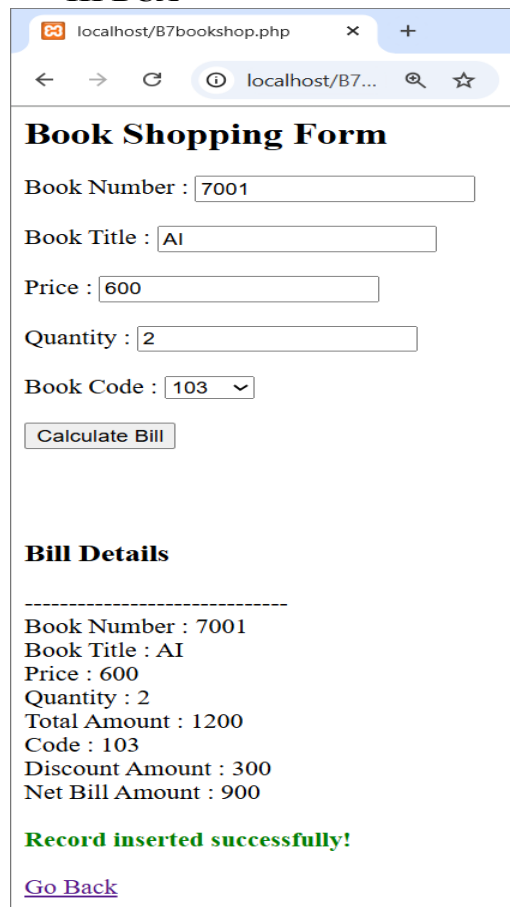
Quantity : 4

Book Code : Other

Calculate Bill

Error: Book Number already exists. Please use a different Book Number.

[Go Back](#)



localhost/B7bookshop.php

Book Shopping Form

Book Number : 7001

Book Title : AI

Price : 600

Quantity : 2

Book Code : 103

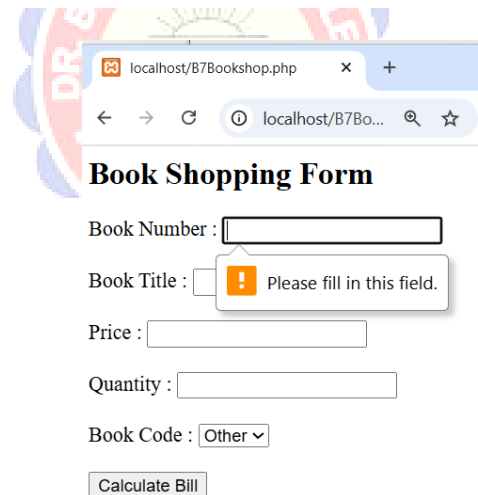
Calculate Bill

Bill Details

Book Number : 7001
Book Title : AI
Price : 600
Quantity : 2
Total Amount : 1200
Code : 103
Discount Amount : 300
Net Bill Amount : 900

Record inserted successfully!

[Go Back](#)



localhost/B7Bookshop.php

Book Shopping Form

Book Number :

Book Title : Please fill in this field.

Price :

Quantity :

Book Code : Other

Calculate Bill

8. Develop a web application for proficiently managing hotel reservations, employing PHP for backend logic and MySQL for data storage. The application should feature a well-structured database table encompassing essential fields such as Room Number (primary key), Room Type (e.g., single semi, single deluxe, double semi, double deluxe, dormitory), Capacity, and Status (booked or available). Within the application's user interface, provide a textbox for entering the room number and two distinct buttons for check-in and check-out functionalities. Ensure that both check in and check-out operations are executed based on the entered room number, facilitating a seamless and intuitive user experience

A. Insert 5 records into the table through interface, reflecting both available and booked rooms.

B. Lists all available rooms and booked rooms on the webpage.

C. Change the booking status to "booked" when a user checks in.

D. Change the room status to "available" when a user checks out.

E. Displays appropriate messages for successful booking, check-out, or if the room number is not present or not in the expected status.

Database Name: BBHC

Table: room

#	Name	Type
1	room_number 🔑	varchar(30)
2	room_type	varchar(30)
3	capacity	varchar(30)
4	status	varchar(30)

Code:

```
<?php
```

```
$conn = new mysqli("localhost", "root", "", "BBHC", 3306);
```

```
if ($conn->connect_error) die("Connection failed: " . $conn->connect_error);
```

```
function execQuery($sql, $types = "", $params = []) {
```

```
    global $conn;
```

```
    $stmt = $conn->prepare($sql);
```

```
if (!$stmt) {  
    die("SQL Prepare Error: " . $conn->error . "<br>Query: " . $sql);  
}  
  
if ($types && $params) {  
    $stmt->bind_param($types, ...$params);  
}  
  
if (!$stmt->execute()) {  
    die("Execution Error: " . $stmt->error);  
}  
  
return $stmt;  
}  
  
function addRecord($num, $type, $cap, $status) {  
    execQuery("INSERT INTO rooms (room_number, room_type, capacity, status) VALUES (?, ?, ?, ?)",  
    "is", [$num, $type, $cap, $status]);  
    echo "Record added successfully.<br>";  
}  
  
function listRooms($status) {  
    global $conn;  
    $res = execQuery("SELECT * FROM rooms WHERE status = ?", "s", [$status])->get_result();  
    echo "<h3>" . ucfirst($status) . " Rooms:</h3>";  
    if ($res->num_rows) {  
        while ($r = $res->fetch_assoc()) {  
            echo "Room: {$r['room_number']}, Type: {$r['room_type']}, Capacity: {$r['capacity']}<br>";  
        }  
    } else {  
        echo "No {$status} rooms.<br>";  
    }  
}
```

```
function updateStatus($room, $status) {
```

```
$stmt = execQuery("UPDATE rooms SET status = ? WHERE room_number = ?", "ss", [$status, $room]);  
echo ($stmt->affected_rows > 0) ? ucfirst($status) . " Successful for Room no $room.<br>" :  
"Room no $room is not found or already $status.<br>" ;  
}
```

```
if ($_SERVER["REQUEST_METHOD"] == "POST") {  
    $room = (int)$_POST["room_number"];  
    if (isset($_POST["add_record"])) {  
        addRecord($room, $_POST["room_type"], (int)$_POST["capacity"], $_POST["status"]);  
    } elseif (isset($_POST["check_in"])) {  
        updateStatus($room, "booked");  
    } elseif (isset($_POST["check_out"])) {  
        updateStatus($room, "available");  
    }  
}
```

```
?>
```

```
<html>
```

```
<head>
```

```
    <title>Hotel Reservation</title>
```

```
</head>
```

```
<body>
```

```
    <h2>Add Room / Alter Room</h2>
```

```
    <form method="post">
```

```
        Room Number: <input type="number" min=1 max=10 name="room_number" required><br><br>
```

```
        Room Type:
```

```
        <select name="room_type" required>
```

```
            <option value="single semi">single semi</option>
```

```
            <option value="single deluxe">single deluxe</option>
```

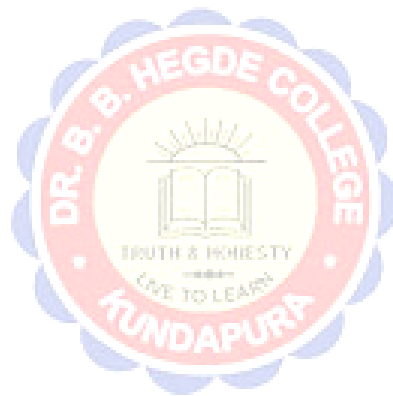
```
            <option value="double semi">double semi</option>
```

```
            <option value="double deluxe">double deluxe</option>
```

```
            <option value="dormitory">dormitory</option>
```

```
        </select><br><br>
```

```
        Capacity: <input type="number" min=1 max=10 name="capacity" required><br><br>
```



Status:

```
<select name="status">
```

```
    <option value="available">available</option>
```

```
    <option value="booked">booked</option>
```

```
</select><br><br>
```

```
<input type="submit" name="add_record" value="AddNewRoom">
```

```
</form>
```

```
<h2>Check-in / Check-out</h2>
```

```
<form method="post">
```

```
    Room Number: <input type="number" min=1 max=10 name="room_number" required><br><br>
```

```
    <input type="submit" name="check_in" value="Checkin">
```

```
    <input type="submit" name="check_out" value="Checkout">
```

```
</form>
```

```
<?php
```

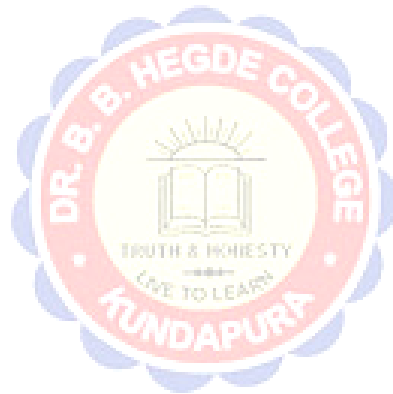
```
listRooms("available");
```

```
listRooms("booked");
```

```
?>
```

```
</body>
```

```
</html>
```



```
<?php
```

```
$conn->close();
```

```
?>
```

Output:**Adding new Room**

Booked successful for Room 1.

Add Room / Alter Room

Room Number:

Room Type: ▼

Capacity:

Status: ▼

AddNewRoom

Check-in / Check-out

Room Number:

Checkin

Checkout

Available Rooms:

Room: 4, Type: single semi, Capacity: 7

Booked Rooms:

Room: 1, Type: double semi, Capacity: 2

Room: 2, Type: single semi, Capacity: 5

Room: 3, Type: single semi, Capacity: 4

Adding new Room

Record added successfully.

Add Room / Alter Room

Room Number:

Room Type: ▼

Capacity:

Status: ▼

AddNewRoom

Check-in / Check-out

Room Number:

Checkin

Checkout

Available Rooms:

Room: 1, Type: double semi, Capacity: 2

Room: 4, Type: single semi, Capacity: 7

Booked Rooms:

Room: 2, Type: single semi, Capacity: 5

Room: 3, Type: single semi, Capacity: 4

Booking for not available room

Room 1 not found or already booked.

Add Room / Alter Room

Room Number:

Room Type:

Capacity:

Status:

AddNewRoom

Check-in / Check-out

Room Number:

Checkin

Checkout

Available Rooms:

Room: 4, Type: single semi, Capacity: 7

Booked Rooms:

Room: 1, Type: double semi, Capacity: 2

Room: 2, Type: single semi, Capacity: 5

Room: 3, Type: single semi, Capacity: 4

Check-Out the Room

Available successful for Room 1.

Add Room / Alter Room

Room Number:

Room Type:

Capacity:

Status:

AddNewRoom

Check-in / Check-out

Room Number:

Checkin

Checkout

Available Rooms:

Room: 1, Type: double semi, Capacity: 2

Room: 4, Type: single semi, Capacity: 7

Booked Rooms:

Room: 2, Type: single semi, Capacity: 5

Room: 3, Type: single semi, Capacity: 4